

Math 118 – Fourier Analysis

Comprehensive exercise solutions

Bogges & Narcowich, *A First Course in Wavelets with Fourier Analysis* (2nd ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{1.2.4} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

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Chapter 0 Inner Product Spaces

Bogges & Narcowich: inner products, projections, Gram–Schmidt, adjoints, least squares

Chapter 1 Fourier Series

Bogges & Narcowich: computation of series, convergence theorems (pointwise, uniform, in mean)

Chapter 2 The Fourier Transform

Bogges & Narcowich: inversion, properties, convolutions, filters, sampling, uncertainty

Chapter 3 Discrete Fourier Analysis

Bogges & Narcowich: DFT, FFT, discrete signals and filters

Chapter 4 Haar Wavelet Analysis

Bogges & Narcowich: the Haar system, decomposition and reconstruction

Chapter 5 Multiresolution Analysis

Bogges & Narcowich: scaling functions, wavelet bases, approximation

Chapter 6 The Daubechies Wavelets

Bogges & Narcowich: construction, moments, smoothness, computation

Chapter 7 Other Wavelet Topics

Boggess & Narcowich: wavelet packets, 2D wavelets, the wavelet transform